



VITALINK TELEHEALTH SOLUTIONS

- VITASTAT ANALYTICS
- Remote Patient Monitoring Optimisation
- Analysing Tool: Tableau



Business Overview

The organization provides remote patient monitoring services for individual with chronic health conditions. Thousands of vital sign readings are generated daily, producing clinical alerts that require timely review by healthcare professionals.

This analysis evaluates operational performance, clinician responsiveness, alert trends, and patient risk to identify opportunities for improving patient safety and efficiency.

Business Objectives



The analysis aims to:



Monitor Clinical alert Volume



Measure clinician response performance



Evaluate compliance with the 2 hour response target



Identify high risk patient groups



Analyse alert severity distribution



Identify regional variations in alert activity



Support evidence based operational decision making

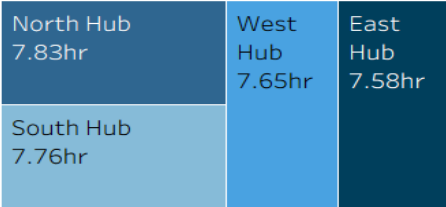
Executive Dashboard

VITALINK TELEHEALTH SOLUTIONS Clinical Response Performance Dashboard

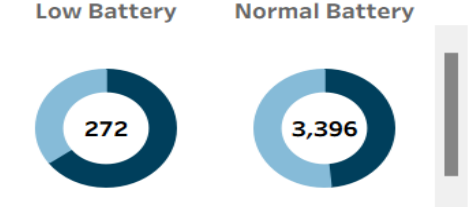
OVERVIEW | WELCOME USER

Total Patients	Total Alert ID	Avg. Resp. Time	Compl. Rate	Breached Alerts
4,417 -	10,500	7.7Hrs -	19.8%	8,425 -

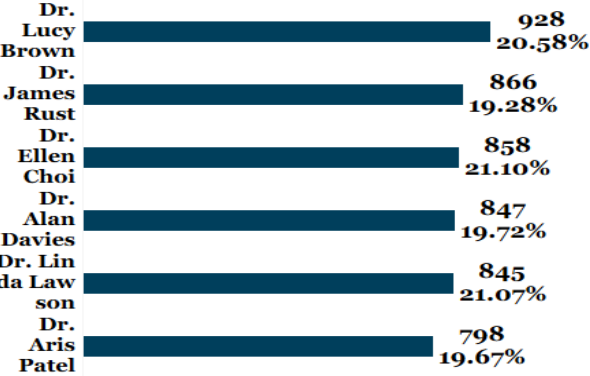
Average Clinician Response Time by Regional Hub



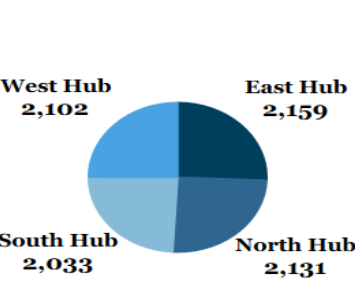
Warning Alerts by Battery Percentage and Connectivity



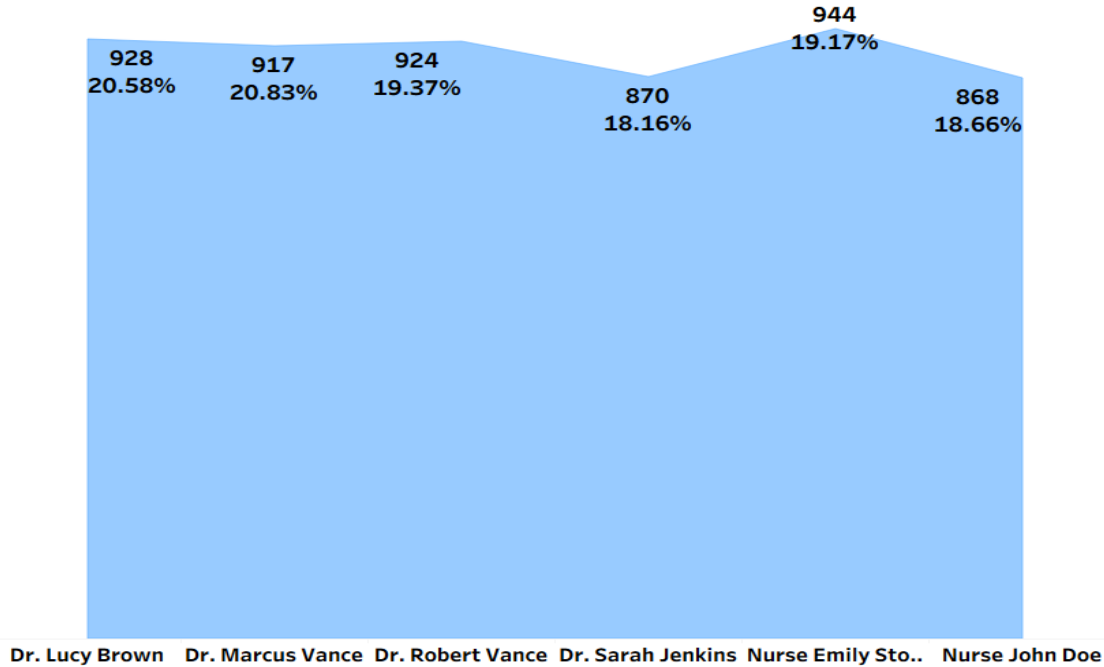
Clinician Alert Volume and Compliance Rate



Regional Hub by Alert Breached



Top 6 Clinician Alerts Volume / Compliance Rate



Executive Dashboard

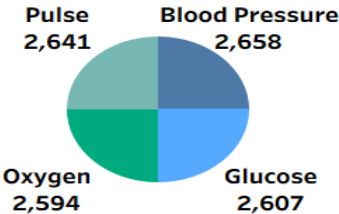
VITALINK TELEHEALTH SOLUTIONS Patient Safety Device Monitoring Dashboard

Youngest Age	Oldest Age	Critical Alert %	Total Warning Alert	Total Alert ID
45	88	15.15%	3,668	10,500

Critical Alert Distribution by Patient City

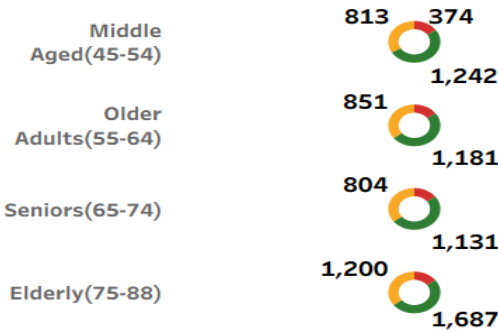


System Alerts by Vital Sign

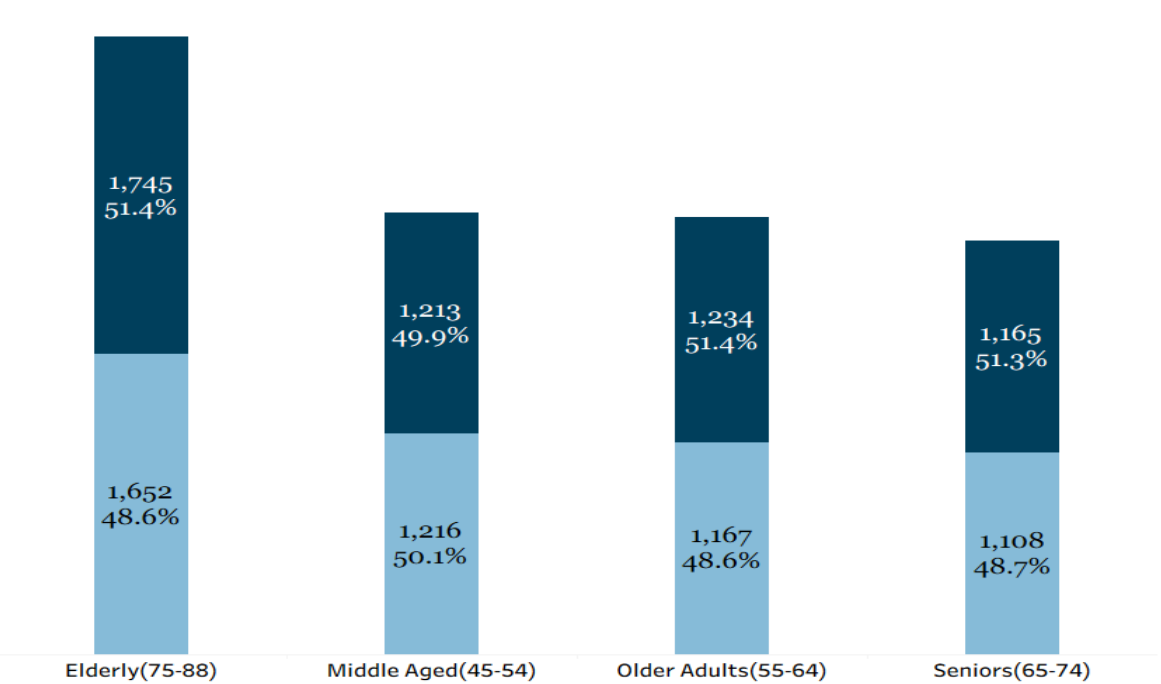


Alert S.. Critical Normal Warning

Patient Age Group by Alert Severity



Alerts Distribution by Age Group and Primary Condition



Clinical Response Performance KPIs

Total Patients	Total Alert ID	Avg. Resp. Time	Compl. Rate	Breached Alerts
4,417 –	10,500	7.7Hrs –	19.8%	8,425 –

Insights

- Total Patients 4,417:** Indicates the overall patient population monitored during the analysis period and provide context for the scale of the service.
- Total Alert ID 10,500:** A high volume of alerts compared with the Patient population indicates a substantial monitoring workload for clinical teams.
- Average Response Time 7.7hrs:** The average response time is significantly above the 2 hours approved target, indicating a major gap in clinical response performance.
- Compliance Rate 19.8%:** Only approximately 1 in 5 alerts met the required response standard, highlighting a significant opportunity to improve clinical responsiveness.
- Breached Alerts 8,425:** A very high number of alerts breached the required response timeframe, indicating that delayed responses are a major operational concern.

Patients Safety Device Monitoring KPIs

Youngest Age	Oldest Age	Critical Alert %	Total Warning Alert	Total Alert ID
45	88	15.15%	3,668	10,500

Insights

- Youngest Age 45:** The monitored population ranges from middle aged to elderly patients, indicating a predominantly higher risk adult population.
- Oldest Age 88:** The inclusion of patients up to age 88 highlights the importance of reliable monitoring and timely intervention for older patients.
- Critical Alerts 15.15%:** Approximately 15% of all alerts are critical, indicating a meaningful proportion of alerts requiring urgent clinical attention.
- Total Warning Alerts 3,668:** Warning alerts represents a substantial monitoring workload and may provide opportunities for earlier intervention before conditions escalate.
- Total Alert ID 10,500:** Confirms the overall volume of alerts generated and provides a baseline for assessing alert burden and system performance.

System Alerts by Vital Signs

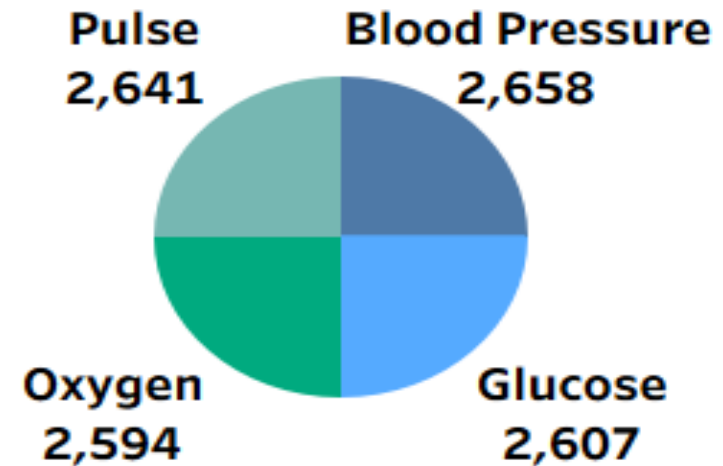
Insight

Blood pressure triggers the highest number of system alerts, accounting for 2,658 alerts (25.3% of all alerts). This is followed closely by Pulse (25.2%), Glucose (24.8%) and Oxygen (24.7%). While the distribution is fairly balanced across all four vital signs, blood pressure generates the highest proportion of alerts, indicating it is the most frequent source of abnormal readings requiring clinical attention.

Recommendation

Prioritise monitoring and early intervention for patients with abnormal blood pressure readings

Review the alert thresholds and clinical protocols to ensure alerts are clinically meaningful and to reduce unnecessary alert fatigue while maintaining patients safety.



Patients Age Group by Alerts Severity

Insight

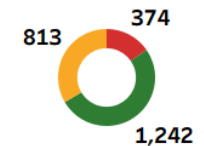
The Elderly (75-88) group has the highest overall alert volume, with 3,397 alerts. However, only 510 (15%) are critical alerts, while the majority are normal alerts.

Recommendation

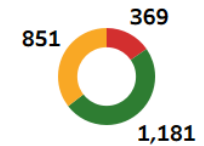
Prioritise monitoring of elderly patients due to their high overall alert volume, while strengthening early warning and escalation processes to ensure critical alerts receive timely clinical attention.

Optimise staffing and allocating additional clinical monitoring capacity during periods when patients generate higher alert volumes helps reduce response delays.

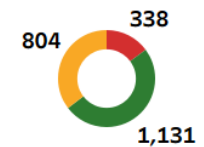
Middle
Aged(45-54)



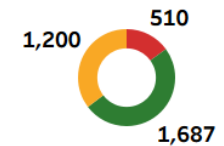
Older
Adults(55-64)



Seniors(65-74)



Elderly(75-88)



Average Clinician Response Time by Regional Hub

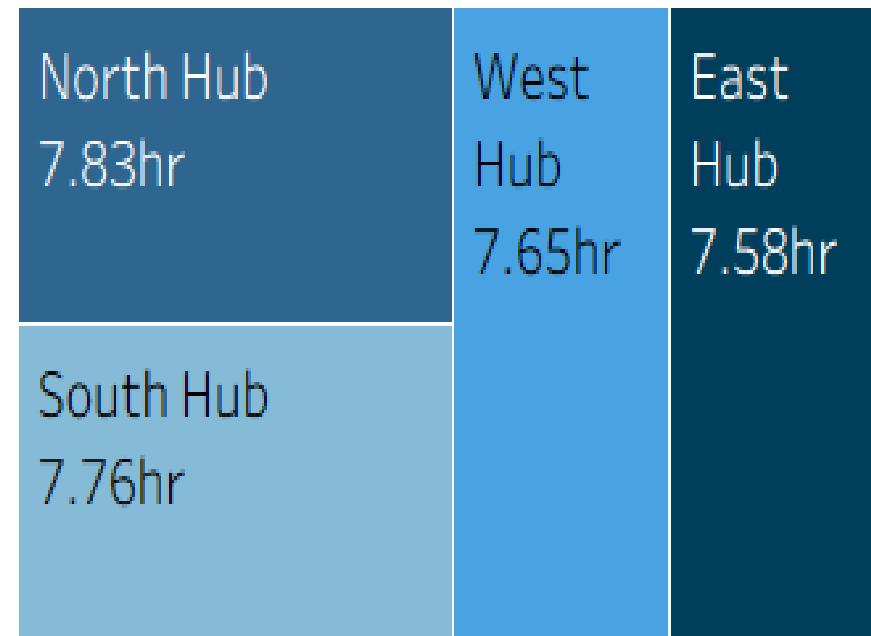
Insight

All four regional hubs substantially exceed the approved 2 hours response time target. The North Hub records the longest average response time at 7.83hrs, followed by the South Hub at 7.76hrs, West Hub at 7.65hrs, and East Hub at 7.58hrs. This indicates a consistent response time gap across all regions rather than an isolated issue at one hub. This indicates a significant performance concern and system wide issue. Even the best performing hub (East) averages 7.58hrs, which is more than 3.5 times the approved 2 hours target.

Recommendation

Management should investigate the underlying causes of the delays, including clinician capacity, alert workload, staffing levels and escalation processes.

Priority should be given to all hubs by introducing automated escalation and workload monitoring to support faster responses and bring performance closer to the 2 hours target.



Regional Hub by Alert Breached

Insight

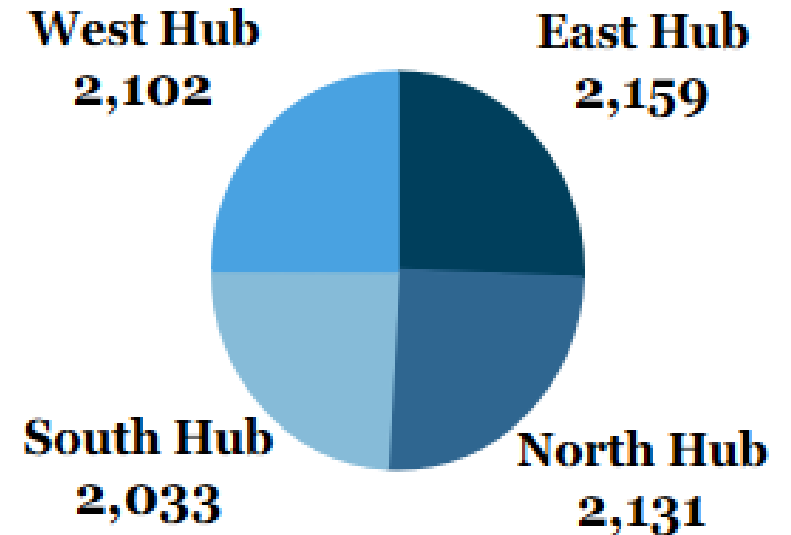
The uniform distribution across the West (2,102), East (2,159), South (2,033), and North (2,131) hubs indicates that alert breaches are not isolated to a single poorly performing region. The cumulative total of 8,425 breaches suggested a widespread operational or technical bottleneck across the entire hubs.

Recommendation

Standardize operational procedure by auditing the workflows, clinician training, or software configurations shared by all hubs to identify root system vulnerabilities.

Evaluate if the alert triggers are too sensitive, which might be causing universal alert fatigue across all hubs.

Introduce automated routing for aging alerts to resolve them before they breach across any hub.



Alert Distribution by Age Group and Primary Conditions

Insight

System alerts are highest among the Elderly age group, accounting for 3,397 alerts (32.4% of all alerts). This is followed by the Middle-aged group with 23.1%, Older Adults with 22.9%, and Seniors with 21.6%. The findings suggest that patients aged 75-88 require the greatest level of clinical monitoring due to the higher frequency of abnormal vital sign alerts.

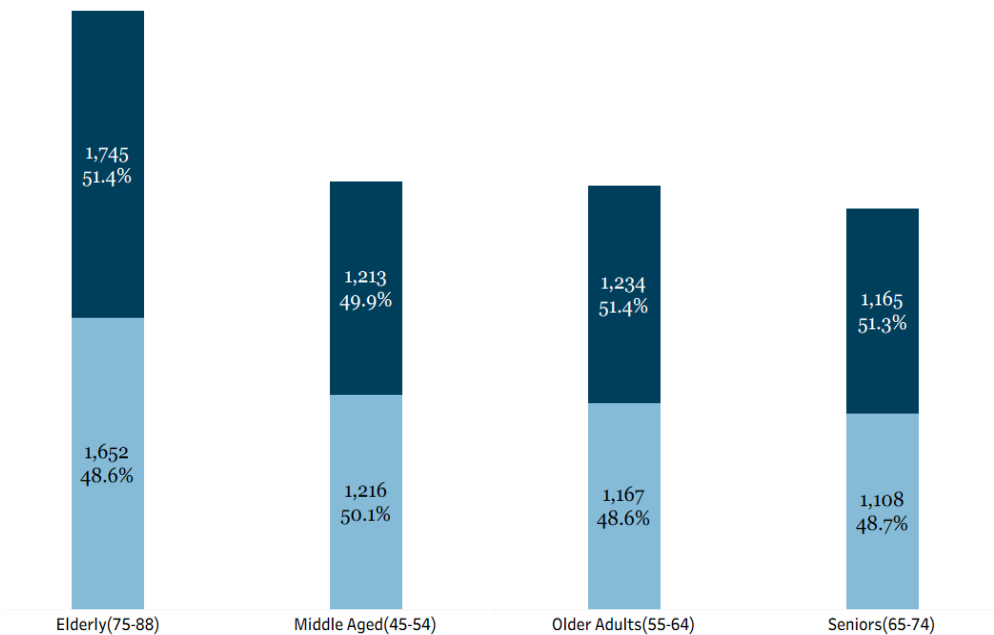
Across all age groups, Diabetes generated 5,357 alerts (51.0%), while Hypertension generated 5,143 alerts (49.0%). The difference is relatively small, indicating that both conditions contribute almost equally to the overall alert burden, with diabetes showing a slight predominance.

Recommendations

Prioritise proactive monitoring and preventive care for elderly patients as they generate nearly one-third of the system alerts.

Implement targeted disease management programmes for both Diabetes and Hypertension, including routine monitoring, medication adherence support, and early intervention for abnormal vital signs.

Healthcare resources and monitoring strategies should address the conditions equally while giving additional attention to high-risk elderly patients.



Critical Alerts Distribution by Patient City

Insight

New York recorded the highest number of critical alerts (418), representing 26.3% of all critical alerts, closely followed by Chicago (413, 26.0%). Boston and Miami accounted for 24.1% and 23.6% respectively. The relatively small differences indicate that critical alerts are evenly distributed across the four cities, with no single city contributing a disproportionate high volume.

Recommendations

Monitoring systems, staffing, and emergency response should be adequately resourced in every city rather than concentrating efforts in one location.

Investigate the slightly higher alert volume in New York and Chicago to determine whether this is caused by a larger patient population, prevalence of chronic conditions or other operational factors.

Preventive care initiatives such as routine health assessments, patient education, medication adherence programmes and remote patient monitoring could help reduce the number of critical alerts.

Tracking critical alerts periodically to identify changes in patient health patterns enables proactive response rather than reactive.



Top 6 Clinicians Alert Volume / Compliance Rate

Insight

Clinician compliance rates are consistently low, ranging from 19.3% - 21.1%. Dr. Ellen Choi achieved the highest compliance rate (21.1%), while Dr. Linda Lawson (21.1%), while Dr. James Rust recorded the lowest (19.3%). The small variation between the clinicians suggests that delayed responses are likely driven by broader operational factors rather than individual performance differences.

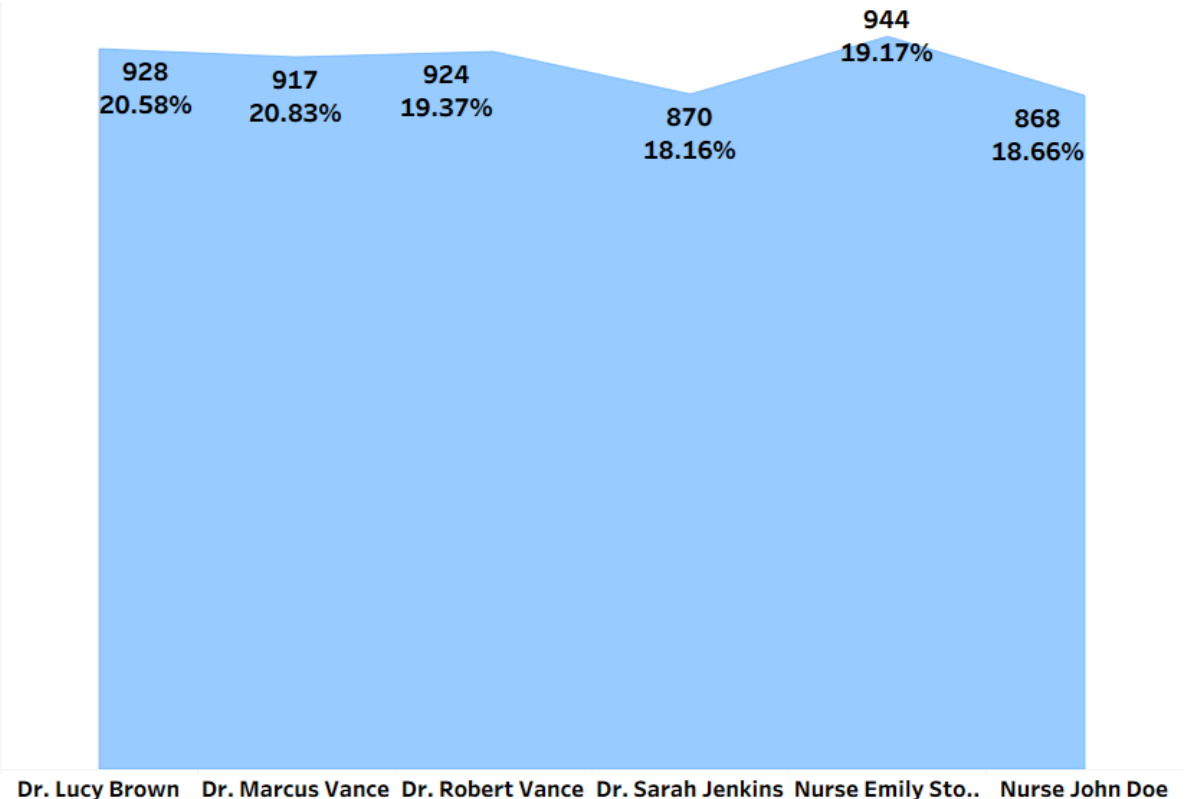
Recommendations

Investigate factors contributing to the low compliance rate such as alert prioritization, staffing and workflow efficiency.

Introduce real time monitoring and escalation for alerts approaching the 2 hours response threshold.

Monitor clinician performance alongside alert volume to ensure workload remains balanced while improving compliance.

Share best practices from clinicians with slightly higher compliance rates to encourage continuous improvement.



Clinicians Warning Alerts by Battery Percentage / Connectivity

Insight

Warning alerts are substantially higher among device with normal battery levels (3,396) compared with low battery devices (272). Within both battery categories, Cellular Connected devices generate slightly more warning alerts than Bluetooth devices. Low battery alerts therefore represent a relatively small proportion of the overall warning alert volume.

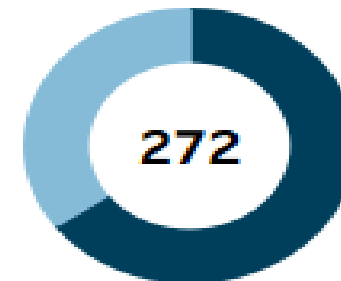
Recommendation

Maintain proactive device battery monitoring and introduce early notifications for devices approaching low battery levels to minimize the risk of monitoring interruptions.

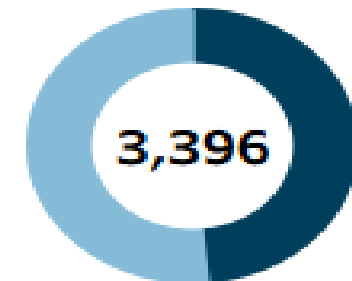
However, because most warning alerts occur when battery levels are normal, further investigation should focus on other potential contributors to warning alerts, such as connectivity performance, device configuration.

Warning Alerts by Battery Percentage and Connectivity

Low Battery



Normal Battery



Clinical Insights Summary

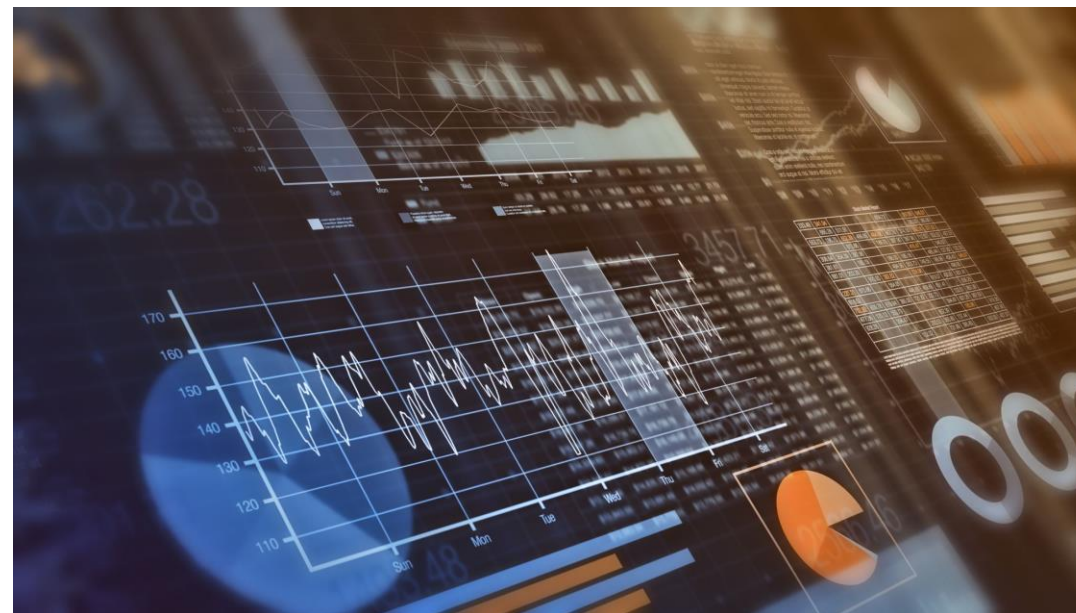
Some Clinicians managed substantially higher alert volumes.

New York generated more critical alerts.

Older patients experienced higher frequencies of clinical alerts.

Blood Pressure monitoring contributed the largest share of escalated alerts.

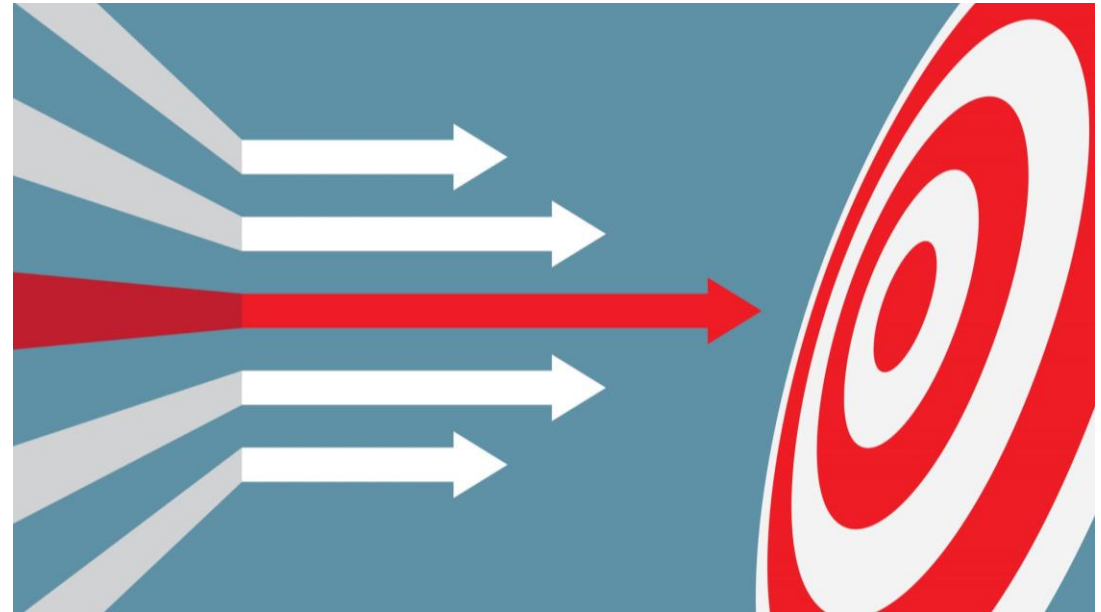
Device connectivity issues were associated with increased warning alerts.



Business Impact

This dashboard enables leadership to:

- ✓ Improve patient safety
- ✓ Reduce response delays
- ✓ Monitor clinician performance
- ✓ Support data driven strategic decision making
- ✓ Improve resource allocation
- ✓ Monitor operational KPIs in real time
- ✓ Identify emerging patient risks
- ✓ Improve clinician accountability



Recommendations Summary



Improve Response
Performance: Introduce
automated escalation for
overdue alerts.



Optimise Workforce
Planning: Redistribute
clinician workload across
operating regions



Strengthen Preventive Care:
Prioritise high risks patients
for proactive monitoring



Improve Device Management:
Monitor battery performance
ans connectivity issues
proactively



Performance Monitoring:
Deploy live operational
dashboards for managers.

Project Reflection

Skills Demonstrated	Healthcare Analysis	Business Intelligence	Data Cleaning	Dashboard Development
KPI Reporting	Dashboard Development	KPI Reporting	Data Storytelling	Operational Analysis
	Performance Monitoring	Tableau Visualisation	Analytical Thinking	

Questions



Thank You



Questions and
Discussion